

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of BSc (MI) Examination April- May 2019

Course code & Name: MI 406 Microbial Physiology

Date: 1/05/2019 Day: Wednesday Time: 1:30 pm To 4:30 pm Maximum Marks: 70

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Part A

Time: 1:30pm To 2:00 pm

Maximum Marks: 20

Q - I Choose the correct answer for the following questions.

20

1.	The following hydrophobic molecule can move around in the cytoplasmic membrane during electron transport (i) cytochrome a (ii) cytochrome b (iii) cytochrome c (iv) Quinones
2.	If the G_f^0 for the formation of a compound is negative then it is formed due to the _____ reaction (i) spontaneous (ii) exothermic (iii) endothermic (iv) any one of the above
3.	The following prosthetic group of flavoproteins is reduced by accepting $2e^-$ from NADH (i) riboflavin (ii) monophosphate (iii) adenine (iv) all of the above
4.	Organisms able to use inorganic chemicals as electron donors are called _____. (i) chemoheterotrophs (ii) chemolithotrophs (iii) phototrophs (iv) heterotrophs
5.	Light-mediated ATP synthesis is called _____. (i) photosynthesis (ii) biosynthesis (iii) photophosphorylation (iv) none of the above
6.	When NH_3 is present at high levels, the following enzyme converts α - ketoglutarate to glutamate (i) glutamate synthase (ii) glutamate dehydrogenase (iii) glutamine synthase (iv) glutamine dehydrogenase

P.T.O.

7.	The following enzyme converts ribose into deoxyribose by reduction of the hydroxyl (-OH) group on the 2' carbon of the 5-carbon pentose ring (i) ribonucleotide reductase (ii) deoxyribonucleotide reductase (iii) ribonucleotide oxidase (iv) deoxyribonucleotide oxidase
8.	The following siderophore are responsible for transport of Fe^{+3} in microbes (i) enterobactins (ii) aquachelin (iii) hydroxamic acids (iv) phenolic
9.	The following organism ferments ethanol plus acetate produced by other organisms (i) <i>Clostridium tetani</i> (ii) <i>Clostridium sporogenes</i> (iii) <i>Clostridium kluyveri</i> (iv) All of the above
10.	Nitrogen fixation is inhibited by oxygen (O_2) because _____ is irreversibly inactivated by O_2 (i) dinitrogenase reductase (ii) phosphoribulokinase (iii) glutamine dehydrogenase (iv) all of the above
11.	Upon a shift from 30° to 42°C , <i>E. coli</i> and other bacteria transiently increase the rate of synthesis of _____ (i) HSP (ii) UreI (iii) SoxR (iv) SoxS
12.	Green plants, algae, and cyanobacteria use electrons from _____ as reducing power during oxygenic photosynthesis (i) CO_2 (ii) H_2O (iii) H_2S (iv) glucose
13.	The _____ participate directly in reactions where light energy is converted into ATP (i) reaction center (ii) antenna pigment (iii) Phycobilin (iv) all of the above
14.	The following chemolithotrophic bacterium deposit elemental sulfur inside the cell, where the sulfur exists as a potential energy (electron) reserve. (i) <i>Ralstonia eutropha</i> (ii) <i>Beggiatoa mirabilis</i> . (iii) <i>Acidithiobacillus thiooxidans</i> (iv) both (i) and (ii)
15.	Active Fnr dimer inactivates the following (i) <i>cyoABCDE</i> (ii) <i>cydAD</i> (iii) <i>narZYWV</i> (iv) all of the above
16.	Following histidine kinase senses low oxygen in the environment (i) ArcA (ii) NarQ (iii) Fnr (iv) EnvZ
17.	The _____ anaerobic bacterium produces endospore. (i) <i>Bacillus anthracis</i> (ii) <i>Clostridium botulinum</i> (iii) <i>B. subtilis</i> (iv) <i>Sporosarcina pasteurii</i>
18.	The _____ is present in high concentration at the poles of the cell, which prevent Z ring formation at the poles of the cell during division. (i) MinC (ii) FtsZ (iii) Zip A (iv) MerB
19.	C- signaling at high intensity induces the _____ in myxobacteria (i) aggregation (ii) sporulation (iii) rippling (iv) all of the above
20.	CtrA-P represses the transcription and prevents initiation of replication in the _____ of caulobacter (i) swarmer cell (ii) stalked cell (iii) both (i) and (ii) (iv) none of the above

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Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

Part B

Time: 2:00pm To 4:30 pm

Maximum Marks: 50

	SECTION -I	20
Q – II A	Discuss briefly any four of the following	(08)
(i)	Regulation of OmpC and OmpF in response to change in osmolality	
(ii)	Formate Nitrate Regulation in response to anaerobic stress	
(iii)	Characteristics of starved cells	
(iv)	Oxidative Stress Response in <i>E. coli</i>	
(v)	Characteristics of thermophile prokaryotes	
Q – II B	Discuss briefly any four of the following	(12)
(i)	α -ketoglutarate triggered heterocyst formation in filamentous cyanobacteria	
(ii)	Life Cycle of <i>Caulobacter crescentus</i>	
(iii)	Intercellular signaling and gliding motility in <i>Myxococcus xanthus</i>	
(iv)	Activation of σ^F during early sporulation	
(v)	Role of Min proteins in septation during cell division	
	SECTION -II	30
Q – III A	Attempt any four of the following:	(08)
(i)	Define glycolysis. Discuss briefly the preparatory stage of glycolysis	
(ii)	Explain the role of TCA cycle in energy generation and biosynthesis	
(iii)	Explain briefly: Redox Reactions giving suitable example	
(iv)	List the types of siderophores and explain its role in mineral uptake	
(v)	Define: Bioenergetics and Free energy	
Q – III B	Explain Electron transport chain in <i>Escherichia coli</i> and its role in generation of proton motive force.	(04)
	OR	
Q – III B	Define energy rich compounds with examples. Explain the role of ATP as an energy rich compound in <i>E.coli</i>	
Q – IV A	Discuss briefly any two of the following	(06)
(i)	Chemoheterotrophic and chemolithotrophic mode of catabolism	
(ii)	Photosynthetic reaction centers and photosynthetic membranes in anoxygenic bacteria with suitable diagram	
(iii)	Lactic Acid Fermentation	P.T.O

Q – IV B	Attempt any three of the following	(06)
(i)	What is Stickland reaction? Explain with suitable example.	
(ii)	Differentiate between: Enriched media and Enrichment media.	
(iii)	Succinate fermentation by <i>Propionigenium modestum</i>	
(iv)	Write a short note on: Mixed acid fermentation	
Q – IV C	Justify any three of the following:	(06)
(i)	Justify: Pigment diversity allows successful coexistence of different phototrophs in the same habitat with suitable examples	
(ii)	Discuss briefly: Electron Flow in Nitrogen Fixation	
(iii)	Discuss briefly: Electron acceptors used in anaerobic respiration	
(iv)	Fastidious microorganisms require enriched medium for growth	